

Erdős Problem #685

EP-685

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $\epsilon > 0$ and n be large depending on ϵ . Is it true that for all $n^\epsilon < k \leq n^{1-\epsilon}$ the number of distinct prime divisors of $\binom{n}{k}$ is $(1 + o(1))k \sum_{k < p < n} \frac{1}{p}$? Or perhaps even when $k \geq (\log n)^c$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>